Coefficient Propagation and Conjugate Obstructions for Trinomial Factors of Newman Polynomials

Local certificates, arithmetic norm divisors, and degree obstructions

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Abstract

The unfair 0–1 polynomial conjecture asks whether monic polynomials with nonnegative real coefficients must themselves be 0–1 polynomials whenever their product is a 0–1 polynomial. In a normalized nonnegative factorization of a Newman polynomial, we study a prospective trinomial factor $A_a(x) = 1+ax^u+x^v$. Coefficient propagation settles the cases in which the two gaps are comparable by divisibility and forces every genuinely fractional coefficient into $0 < a < 1/2$. Algebraic integrality and the golden-ratio annulus for zeros of Newman polynomials give explicit bounds on all conjugates. The complete norm polynomial is an integer divisor of the product, yielding finite searches at fixed algebraic degree. A unit-circle crossing argument excludes quadratic coefficients when the smaller gap is 2. Finally, an exact rational Sturm–Schur sieve and a parity-block expansion of the forced cofactor recurrence exclude every cubic coefficient in every remaining smaller-gap-2 case, with no product-degree cap. An exact quartic norm sieve and an expanded parity-block certificate also exclude every quartic coefficient, giving under these factorization hypotheses the algebraic-degree lower bound $[\mathbb{Q}(a) : \mathbb{Q}] \geq 5$. The finite cubic catalog has 126 entries. Exact algebraic signs settle $v = 5, 7, \ldots, 21$, while rational interval bounds for alternating binomial tails prove uniform positivity followed by a negative coefficient for every odd $v \geq 23$. For quartics the exact sieve retains 6,977 minimal polynomials and 6,982 target roots; the corresponding negative coefficient occurs no later than degree $110v + 100$. Independently, a unit-disk/Pringsheim argument proves eventual negativity of the forced series for every real $a > 0$. A safe-lattice theorem proves that the only possible coefficient-one escape cannot first occur before degree $13v+4$ at an odd index or $14v+2$ at an even index. These are structural restrictions, not a proof of the full conjecture: the safe-lattice theorem does not rule out a later escape, and the degree-5 and higher cases remain open.

1 Introduction

A *Newman polynomial* has coefficients in $\{0, 1\}$ and constant term 1. We consider the following open problem.

Unfair 0–1 polynomial conjecture. If $A, B \in \mathbb{R}[x]$ are monic with nonnegative coefficients and AB is a 0–1 polynomial, then both A and B are 0–1 polynomials.

The conjecture is equivalent to a factorization question for finitely supported probability measures. The arithmetic-progression case goes back to Krasner and Ranulac [6]. Ghidelli treated the first difficult trinomial $1+ax^2+x^5$ by recurrence and root analysis [1], and Hare developed recursive symbolic methods for bounded-degree candidate factors [2]. The general conjecture remains open.

This paper isolates a uniform local coefficient mechanism and joins it to the arithmetic theory of Newman divisors. Throughout, the factor statements concern normalized factorizations with nonnegative real cofactors. The principal conclusions are:

1. divisibility-comparable gaps force fairness;
2. a fractional middle coefficient is an irrational algebraic integer in $(0, 1/2)$;
3. every conjugate satisfies explicit gap-dependent bounds;
4. the full norm polynomial is an effectively computable integer divisor;
5. smaller gap 2 excludes algebraic degrees 2, 3, and 4 without a product-degree restriction;
6. in the unrestricted smaller-gap-2 recurrence, a first coefficient-one escape is impossible below $13v+4$ at odd indices and $14v+2$ at even indices.

The last assertion is an analytic theorem in the same normalized nonnegative-factorization setting, accompanied by reproducible exact identity and symbolic sign audits. Its artifacts include source code, exact root records, exact algebraic signs, rational interval margins, alternating-tail bounds, and standalone checkers. No priority claim is made for the general propagation or norm-divisor mechanisms.

2 Normalization and the recurrence

Lemma 2.1 (Endpoint normalization)**.** *Let $A, B \in \mathbb{R}[x]$ be nonzero and monic with nonnegative coefficients, and suppose $C = AB$ is a 0–1 polynomial. After removing powers of x from A and B, both constant coefficients are 1, and every coefficient of A and B belongs to $[0, 1]$.*

Proof. Let a_0, b_0 be the first nonzero coefficients after removing the powers of x. The first nonzero coefficient of C is $a_0 b_0 = 1$. The coefficient in degree $\deg B$ contains the nonnegative summand a_0, so $a_0 \le 1$; similarly $b_0 \le 1$. Hence $a_0 = b_0 = 1$. Multiplication by the constant coefficient of the other factor makes each coefficient of A or B a nonnegative summand of a coefficient of C, and hence at most 1. □

If one normalized factor is already a 0–1 polynomial, low-to-high coefficient comparison makes the other factor integral; Lemma 2.1 then makes it a 0–1 polynomial as well.

For

$$A_a(x) = 1 + ax^u + x^v, \qquad 0 < u < v,$$

write $B(x) = \sum_{n\ge 0} b_n x^n$ and $C(x) = \sum_{n\ge 0} c_n x^n$, extending $b_n = 0$ outside the degree of B. Then

$$c_n = b_n + a b_{n-u} + b_{n-v} \in \{0, 1\}, \qquad b_n \ge 0, \tag{2.1}$$

with $b_0 = 1$ and $b_j = 0$ for $j < 0$. Whenever one summand in (2.1) is positive, c_n is forced to be 1.

3 Local propagation along arithmetic rays

Lemma 3.1 (Arithmetic-ray overflow)**.** *Let $r \ge 1$, $k \ge 2$, and $0 < a < 1$. Suppose a polynomial or formal series begins*

$$A(x) = 1 + ax^r + x^{kr} + O(x^{kr+1}),$$

with no other nonzero coefficient through degree kr. There is no series $B = \sum_{j\geq 0} b_j x^j$ with $b_0 = 1$, $b_j \geq 0$, for which all coefficients of AB through degree kr lie in $\{0, 1\}$.

Proof. At degrees $r, 2r, \ldots, (k-1)r$, positivity forces the product coefficient to be 1, and therefore

$$b_{jr} = 1 - ab_{(j-1)r} = 1 - a + a^2 - \cdots + (-a)^j = \frac{1 - (-a)^{j+1}}{1 + a} > 0.$$

At degree kr, the terminal unit term collides with this positive tail: $[x^{kr}](AB) = b_{kr} + ab_{(k-1)r} + b_0 > 1$. □

Theorem 3.2 (Divisible gaps)**.** *Suppose $A_a = 1 + ax^u + x^v$ occurs in a normalized nonnegative factorization of a 0–1 polynomial. If one of u and $v - u$ divides the other, then $a \in \{0, 1\}$ and both factors are 0–1 polynomials.*

Proof. If $u \mid v$, apply Lemma 3.1 with $r = u$. If $(v - u) \mid v$, reverse the factorization using $P^*(x) = x^{\deg P} P(x^{-1})$; the reversed trinomial is $1 + ax^{v-u} + x^v$. These two alternatives are equivalent to comparability of u and $v - u$ by divisibility. Thus $0 < a < 1$ is impossible, and the observation after Lemma 2.1 proves the final claim. □

Theorem 3.3 (Universal half-bias obstruction)**.** *If A_a occurs in a normalized nonnegative factorization of a 0–1 polynomial and $0 < a < 1$, then $a \leq 1/2$.*

Proof. After reversal assume $u \leq v - u$. Equality is covered by Theorem 3.2, so $2u < v$. Recurrence (2.1) gives $b_u = 1 - a$ and $b_{2u} = 1 - a + a^2$. At degree v, the unit contribution from $x^v b_0$ forces $b_v = b_{v-u} = 0$; the next position gives $b_{v+u} = a$. Consequently

$$[x^{v+2u}](A_a B) = b_{v+2u} + ab_{v+u} + b_{2u} = b_{v+2u} + 1 - a + 2a^2.$$

For $a > 1/2$ this exceeds 1, a contradiction. □

Proposition 3.4 (Forced-ray skeleton)**.** *Assume the hypotheses of Theorem 3.3, with $2u < v$. By Theorem 3.2, the feasible fractional case has $u \nmid v$. Thus write $v = qu + r$, where $q \geq 2$ and $0 < r < u$. Put*

$$S_j(a) = \sum_{k=0}^{j} (-a)^k, \qquad T_j(a) = \sum_{k=0}^{j-1} (k+1)(-a)^k.$$

Every feasible cofactor satisfies

$$b_{ju} = S_j(a) \ (0 \leq j \leq q), \quad b_{r+ju} = 0 \ (0 \leq j \leq q-1), \quad b_{v+ju} = aT_j(a) \ (1 \leq j \leq q),$$

and

$$T_j(a) = \frac{1 - (j+1)(-a)^j + j(-a)^{j+1}}{(1+a)^2}.$$

For $0 < a \leq 1/2$, every $T_j(a)$ is nonnegative; for $0 < a < 1/2$ it is positive. Thus this paired ray gives no stronger upper bound before the next interaction at degree $2v$.

Proof. The first identity is the recurrence below degree v. At degree v one has $b_{v-u} = 0$. If a zero $b_{r+ju} = 0$ in this residue class had $b_{r+(j-1)u} > 0$, then the recurrence below degree v would force $ab_{r+(j-1)u} = 1$, hence $b_{r+(j-1)u} = 1/a > 1$, contrary to Lemma 2.1. Backward induction gives the displayed zero chain. Put $d_j = b_{v+ju}$ and $d_0 = 0$. Since $S_j(a) > 0$, the product coefficient at $v + ju$ is 1, whence $d_j = a(S_{j-1}(a) - d_{j-1})$. Induction yields $d_j = aT_j(a)$, and differentiating a finite geometric sum gives the closed form. For $0 < a < 1/2$, the magnitudes $(k+1)a^k$ decrease strictly, so the finite alternating sum is positive. At $a = 1/2$ they are nonincreasing and the only zero case is $T_2(1/2) = 0$; hence nonnegativity holds on the closed interval. □

4 Algebraic integrality and the norm divisor

Theorem 4.1 (Algebraic integrality)**.** *Let $C \in \mathbb{Z}[x]$ be monic and $C = AB$ with $A, B \in \mathbb{R}[x]$ monic. Every coefficient of A and B is an algebraic integer. In particular, a rational coefficient is an integer.*

Proof. Every root of either factor is a root of the monic integer polynomial C and is therefore an algebraic integer. Factor coefficients are elementary symmetric polynomials in these roots. A rational algebraic integer is an ordinary integer. □

Thus a genuinely fractional coefficient satisfies

$$0 < a < \frac{1}{2}, \qquad a \text{ is an irrational algebraic integer.} \tag{4.1}$$

Theorem 4.2 (Conjugate and norm sieve)**.** *Suppose $A_a(x) = 1 + ax^u + x^v$ occurs in a normalized nonnegative factorization $C = A_a B$, where C is a Newman polynomial. Let a have degree d, let $m_a(t)$ be its minimal polynomial, and put $\varphi = (1+\sqrt{5})/2$. For every conjugate α of a:*

(a) $A_\alpha(x) = 1 + \alpha x^u + x^v$ divides C in $\overline{\mathbb{Q}}[x]$;

(b)

$$|\alpha| \le R_\varphi(u,v) := \min\{\varphi^u + \varphi^{u-v},\ \varphi^{v-u} + \varphi^{-u}\}; \tag{4.2}$$

(c) if α is real, then $\alpha > -\kappa(u,v)$, where

$$\kappa(u,v) = \left(\frac{v-u}{u}\right)^{u/v} + \left(\frac{u}{v-u}\right)^{(v-u)/v}. \tag{4.3}$$

Furthermore,

$$H_a(x) := \prod_\alpha (1 + \alpha x^u + x^v) = \operatorname{Res}_t\big(m_a(t), 1 + tx^u + x^v\big) \tag{4.4}$$

belongs to $\mathbb{Z}[x]$, divides C in $\mathbb{Z}[x]$, and has degree dv. Consequently $\deg C \ge dv$.

Proof. Division of C by the monic polynomial A_a takes place in $\mathbb{Q}(a)[x]$. Applying an embedding $\mathbb{Q}(a) \hookrightarrow \mathbb{C}$ to $C = A_a B$ fixes C and proves (a).

Odlyzko and Poonen's zero bound gives

$$\varphi^{-1} < |z| < \varphi \tag{4.5}$$

for every zero of a Newman polynomial [7]. On $|z| = \varphi^{-1}$, the strict inequality $|\alpha|\varphi^{-u} > 1 + \varphi^{-v}$ would make αz^u dominate $1 + z^v$; Rouché's theorem would place u zeros of A_α in the forbidden inner disk. Hence $|\alpha| \le \varphi^u + \varphi^{u-v}$. On $|z| = \varphi$, the analogous strict inequality would make A_α have only u zeros in that circle, whereas all v of its zeros must lie there by (a) and (4.5). This gives the other bound in (4.2).

For (c), write $\alpha = -s$ with $s > 0$. The function $f(x) = 1 - sx^u + x^v$ on $x > 0$ first acquires a positive zero at tangency. Solving $f(x) = f'(x) = 0$ gives $s = \kappa(u,v)$. Thus for $s \ge \kappa(u,v)$, A_{-s} has a positive real zero. Since A_{-s} divides C, that zero would also be a zero of C, impossible because C has nonnegative coefficients and $C(0) = 1$. Notice that no coefficient-sign claim about A_{-s} is used.

Distinct conjugate trinomials are coprime: their difference is a nonzero multiple of x^u, and neither vanishes at 0. Their product is Galois-fixed and has algebraic-integer coefficients, hence lies in $\mathbb{Z}[x]$. By (a) and coprimality it divides C in $\overline{\mathbb{Q}}[x]$, hence in $\mathbb{Q}[x]$; Gauss's lemma gives divisibility in $\mathbb{Z}[x]$. Formula (4.4) and the degree claim follow directly. □

5 Effective finiteness at fixed algebraic degree

Theorem 5.1 (Effective finite enumeration)**.** *Fix u, v and d. There are only finitely many algebraic integers a of degree d that can occur in* (4.1)*. More explicitly, if*

$$m_a(t) = t^d + s_{d-1}t^{d-1} + \cdots + s_0 \in \mathbb{Z}[t],$$

and $R = R_\varphi(u, v)$, then

$$|s_{d-k}| \le \binom{d}{k} R^k, \qquad 1 \le k \le d. \tag{5.1}$$

Every candidate can therefore be enumerated in finite integer boxes and tested against the necessary root and conjugate conditions. For any fixed product-degree cap D, existence of a Newman multiple is a finite integer feasibility problem for the quotient coefficients. If $\deg C \le D$, it is enough to consider $d \le \lfloor D/v \rfloor$.

Proof. Up to sign, s_{d-k} is the kth elementary symmetric function of the d conjugates. It has exactly $\binom{d}{k}$ summands, each of modulus at most R^k, which proves (5.1). The coefficient boxes are finite. For a fixed candidate H_a and cap D, write $C = H_a Q$ with $\deg Q \le D - dv$. Since $H_a(0) = C(0) = 1$, one has $q_0 = 1$, and low-to-high coefficient recursion

$$q_n = c_n - \sum_{i=1}^{n} h_i q_{n-i}$$

shows inductively that every q_n is integral. Equating the finitely many coefficients and requiring each coefficient of C to lie in $\{0, 1\}$ then gives the stated finite integer system. The degree restriction follows from $dv \le \deg C$. □

6 The smaller-gap-two quadratic case

Lemma 6.1. *For every integer $v \ge 5$ one has $\kappa(2, v) < 2$.*

Proof. Put $p = 2/v$ and $r = p/(1-p) = 2/(v-2)$. Then $0 < r \le 2/3$ and

$$\kappa(2, v) = r^{-p} + r^{1-p} = (1+r)r^{-r/(1+r)} =: K(r).$$

Direct differentiation gives

$$\frac{d}{dr} \log K(r) = -\frac{\log r}{(1+r)^2} > 0 \qquad (0 < r < 1).$$

Hence $K(r) \le K(2/3) = \frac{5}{3}(3/2)^{2/5} < 2$. The final inequality is equivalent, after taking fifth powers, to $9/4 < (6/5)^5$, i.e. $28125 < 31104$. □

Proposition 6.2 (Unique coarse quadratic survivor)**.** *Suppose* $\min\{u, v-u\} = 2$*. If a genuinely fractional coefficient a is quadratic, then necessarily $a = (3 - \sqrt{5})/2$.*

Proof. Reverse if necessary, so $u = 2 \le v - u$. Theorem 3.2 removes even v and equality, so $v \ge 5$ is odd. Let a' be the other conjugate. If the integer norm aa' were negative, then $a' \le -1/a < -2$. On the other hand Theorem 4.2(c) and Lemma 6.1 give $a' > -\kappa(2, v) > -2$, a contradiction. Thus $a' > 0$. The modulus bound gives

$$a' \le \varphi^2 + \varphi^{2-v} \le \varphi^2 + \varphi^{-3} = \frac{3\sqrt{5} - 1}{2} < 2.855.$$

Writing the minimal polynomial as $t^2 - Tt + N$, one obtains $0 < N = aa' < 2$ and $0 < T = a + a' < 4$. Hence $N = 1$ and $T \le 3$. The only monic quadratic with a positive root in $(0, 1/2)$ is $t^2 - 3t + 1$. □

Lemma 6.3 (Boundary-root perturbation)**.** *Let $v \geq 5$ be odd and $0 < t < 2\sin(\pi/(v-1))$. Then*

$$g_t(z) = 1 + z^2 + tz^v$$

has exactly two zeros in the open unit disk.

Proof. At $t = 0$ the two roots are $\pm i$. The implicit-function theorem gives an analytic branch from i with

$$z(t) = i - \frac{i^{v-1}}{2}t - \frac{2v-1}{8}it^2 + O(t^3), \qquad |z(t)|^2 = 1 - \frac{v-1}{2}t^2 + O(t^3).$$

Thus this branch and its conjugate branch from $-i$ lie inside the unit circle for all sufficiently small $t > 0$. They are the only bounded branches: if $t_n \to 0$ and zeros z_n remain bounded, every limit point solves $1 + z^2 = 0$; the two simple implicit branches account for all such zeros. Hence, for sufficiently small positive t, exactly two zeros lie in the unit disk.

It remains to rule out boundary crossings. If $g_t(e^{i\theta}) = 0$, then multiplication by $e^{-i\theta}$ gives

$$2\cos\theta + te^{i(v-1)\theta} = 0.$$

Therefore $(v-1)\theta = k\pi$ and necessarily

$$t = -2(-1)^k \cos\left(\frac{k\pi}{v-1}\right) > 0.$$

Because $v - 1$ is even, the zero cosine occurs at $\theta = \pi/2$ modulo π, and the nearest nonzero point of the grid has absolute cosine $\sin(\pi/(v-1))$. Thus no unit-circle zero exists for $0 < t < 2\sin(\pi/(v-1))$. The argument principle now makes the number of zeros in the unit disk constant throughout this interval, so it remains two. □

Theorem 6.4 (Quadratic exclusion)**.** *If $1 + ax^u + x^v$ is a genuinely fractional factor in a normalized nonnegative factorization of a Newman polynomial and $\min\{u, v-u\} = 2$, then $[\mathbb{Q}(a) : \mathbb{Q}] \geq 3$.*

Proof. Assume a is quadratic. Proposition 6.2 gives $a = (3 - \sqrt{5})/2$ and conjugate $a' = (3 + \sqrt{5})/2 = \varphi^2$. Reverse so $u = 2$ and $v \geq 5$ is odd. Under $z = \varphi x$,

$$A_{a'}(z/\varphi) = 1 + z^2 + \varphi^{-v}z^v.$$

Moreover $\varphi^{-v} < 2\sin(\pi/(v-1))$: indeed $2\sin(\pi/(v-1)) \geq 4/(v-1)$, while $\varphi^v > F_v \geq v - 1$ for $v \geq 5$. Lemma 6.3 yields a zero $|z| < 1$, hence a zero $|x| < \varphi^{-1}$ of $A_{a'}$. Theorem 4.2(a) makes this a zero of C, contradicting (4.5). □

7 Global cubic exclusion for the smaller gap two

This section replaces a bounded quotient search by a forced-recurrence certificate. The argument has two exact computational layers. First, the uniform norm sieve produces the same 126 possible cubic minimal polynomials for every odd $v \geq 5$. Second, for the distinguished root $a \in (0, 1/2)$ of each polynomial, a parity-block expansion proves that the cofactor recurrence stays strictly positive, apart from its forced initial zeros, until a specified coefficient becomes negative. The obstruction index is at most $27v + 8$.

Remark 7.1 (Why the nonnegative cofactor is essential). Let $a = 2\cos(4\pi/9)$, the root in $(0, 1/2)$ of the irreducible polynomial $t^3 - 3t + 1$. Exact multiplication, using $a^3 = 3a - 1$, gives

$$1 + x^{18} = (1 + ax^2 + x^4)B(x),$$

where

$$\begin{aligned} B(x) = x^{14} - ax^{12} + (a^2 - 1)x^{10} + (1 - a)x^8 \\ + (1 - a)x^6 + (a^2 - 1)x^4 - ax^2 + 1. \end{aligned}$$

Thus cubic ordinary Newman divisibility does occur. The negative coefficients of B show why the normalized nonnegative-factorization hypothesis cannot be discarded.

Lemma 7.2 (Uniform cubic catalog). *If $A_a(x) = 1 + ax^2 + x^v$ occurs in a normalized nonnegative factorization, $v \geq 5$ is odd, and $[\mathbb{Q}(a) : \mathbb{Q}] = 3$, then the minimal polynomial of a is one of the* 126 *polynomials in the accompanying exact catalog.*

Proof. Write $m_a(t) = t^3 + s_2t^2 + s_1t + s_0$. For every odd $v \geq 5$, Theorem 4.2 gives

$$|\alpha| < \varphi^2 + \varphi^{2-v} \leq \varphi^2 + \varphi^{-3} < \frac{143}{50}$$

for every conjugate α of a. Therefore

$$|s_2| \leq 8, \qquad |s_1| \leq 24, \qquad |s_0| \leq 23.$$

If α, β are the other two conjugates, then $|s_0| = |a\alpha\beta|$ is a positive integer. Hence the target root also has the uniform lower bound

$$a > \left(\frac{50}{143}\right)^2 = \frac{2500}{20449}.$$

The real-conjugate estimate also gives $\alpha > -\kappa(2, v) > -2$ for every real conjugate. The exact generator enumerates all 38318 triples in the boxes with nonzero constant coefficient, removes reducible cubics by the integer-root test, counts roots in $(0, 1/2)$ and $(-3, -2]$ by rational Sturm sequences, and applies rational Schur recursion for the disk bound. Exactly 126 irreducible cubics remain. The independent checker repeats this enumeration without importing the generator. □

For the recurrence certificate, define the formal series

$$Q_{a,v}(x) := \frac{1 + x^v}{(1 - x^2)(1 + ax^2 + x^v)} = \sum_{n \geq 0} q_n x^n. \tag{7.1}$$

The eventual sign change in this series is not special to algebraic degree three.

Lemma 7.3 (Analytic sign change). *For every $a > 0$ and every odd $v \geq 5$, the series $Q_{a,v}$ has a negative coefficient.*

Proof. Put $A(x) = 1 + ax^2 + x^v$. Since A is monic with constant coefficient 1, the product of the moduli of its roots is 1. If A had no root in the open unit disk, all its roots would therefore lie on the unit circle. Reality of the coefficients would then make the root multiset invariant under inversion, so the monic reciprocal $x^vA(x^{-1}) = x^v + ax^{v-2} + 1$ would equal A. This is impossible for $a > 0$ and $v \geq 5$. Thus A has a root of modulus strictly less than 1.

No zero of A cancels against $1 + x^v$: a common zero would give $ax^2 = 0$. Nor does A vanish at 1 or -1. Hence the radius of convergence of $Q_{a,v}$ is some $R < 1$. If every q_n were nonnegative, Pringsheim's theorem would make the positive real point R a singularity. But the displayed rational function has no singularity on $(0, 1)$, since both $1 - x^2$ and $1 + ax^2 + x^v$ are positive there. This contradiction proves the claim. □

Remark 7.4 (General nonpalindromic gaps). The same proof works with 2 replaced by any u with $0 < u < v$ and $2u \neq v$: the formal series

$$\frac{1 + x^v}{(1 - x^u)(1 + ax^u + x^v)}$$

has a negative coefficient for every $a > 0$. Indeed, if all roots of the trinomial lay on the unit circle, reciprocity would force the middle exponents u and $v - u$ to agree. This analytic statement alone does not prove that an arbitrary finite cofactor follows the displayed forced-ray series up to its first negative term.

The numerator is not an arbitrary guess: if the cofactor coefficients remain nonnegative, positivity forces the product coefficients to be 1 on the even ray and on the odd ray beginning at v.

Lemma 7.5 (Forced-series obstruction). *Let $0 < a < 1$, let $v \geq 5$ be odd, and suppose there is an index $N > v$ such that the coefficients of* (7.1) *satisfy*

$$0 \leq q_n < 1 \qquad (0 < n < N),$$

and $q_N < 0$. Then $1 + ax^2 + x^v$ has no finite monic nonnegative cofactor whose product is a Newman polynomial.

Proof. Extend a prospective cofactor $B = \sum b_n x^n$ by zero outside its degree. Endpoint normalization gives $0 \leq b_n \leq 1$ for every n. The coefficient at degree v contains $b_0 = 1$, so it forces $b_v = b_{v-2} = 0$. If $b_j = 0$ at an odd index $j < v$ but $b_{j-2} > 0$, then the coefficient at degree j would force $ab_{j-2} = 1$, and hence $b_{j-2} = 1/a > 1$, a contradiction. Backward propagation along the odd ray therefore gives $b_1 = b_3 = \cdots = b_{v-2} = 0$. Along the even ray below v, positivity forces the product coefficient to be 1 and hence gives the coefficients of $Q_{a,v}$.

For $n \geq v$, the defining recurrence for $Q_{a,v}$ gives

$$aq_{n-2} + q_{n-v} = 1 - q_n.$$

This forcing term is strictly positive for $n < N$ by hypothesis and is greater than 1 at $n = N$. Thus the product coefficient is forced to be 1, and induction gives $b_n = q_n$ through $\min\{N, \deg B\}$. If $\deg B \geq N$, this yields the contradiction $b_N = q_N < 0$. If $0 < \deg B < N$, then the leading coefficient $b_{\deg B} = 1$. But $b_{\deg B} = q_{\deg B} < 1$, again contradicting monicity. Finally $B = 1$ would leave the fractional coefficient a in the product. □

Proposition 7.6 (The coefficient-one escape). *Fix $0 < a < 1$ and odd $v \geq 5$. Suppose that, through degree $n - 1 \geq v$, a prospective nonnegative cofactor agrees with the forced series $Q_{a,v} = \sum q_j x^j$. Put*

$$f_n := aq_{n-2} + q_{n-v}.$$

Then $f_n > 1$ is an immediate contradiction; if $0 < f_n \leq 1$, the Newman product coefficient at degree n is forced to be 1 and the next cofactor coefficient is forced to be $q_n = 1 - f_n$; and if $f_n = 0$, the first deviation from the forced series is possible. In the last case $q_n = 1$, and, while the preceding coefficients are nonnegative,

$$f_n = 0 \quad \Longleftrightarrow \quad q_{n-2} = q_{n-v} = 0.$$

No such simultaneous predecessor-zero event occurs for $v < n \leq 2v$.

Proof. At every degree $n \geq v$, the target ray series has coefficient 1, so $q_n = 1 - f_n$. If the actual product coefficient is $c_n \in \{0, 1\}$, then nonnegativity of the next cofactor coefficient gives $b_n = c_n - f_n \geq 0$. Thus $f_n > 0$ forces $c_n = 1$, and gives the first two alternatives; when $f_n = 0$, either $c_n = 1$ and $b_n = q_n = 1$, or $c_n = 0$ and $b_n = 0$, which is the first possible branch away from $Q_{a,v}$. Because $a > 0$, a zero sum of the two nonnegative predecessors is equivalent to their simultaneous vanishing.

For $v < n \leq 2v$, if n is even then q_{n-v} lies on the initial odd zero ray whereas $q_{n-2} = 1 - a + a^2 - \cdots$ is strictly positive for $0 < a < 1$. If n is odd, then q_{n-v} is such a positive alternating geometric sum. Hence $f_n > 0$ throughout this range. □

Lemma 7.7 (Parity rationalization)**.** *Put $z = x^2$ and define $r_m = r_m(t)$ by*

$$R_{t,v}(z) := \sum_{m \geq 0} r_m z^m = \frac{1}{(1-z)((1+tz)^2 - z^v)}, \qquad r_m = 0 \quad (m < 0).$$

Then the forced-series coefficients, regarded as polynomials in t, satisfy

$$\begin{aligned} q_{2m}(t) &= r_m + t r_{m-1} - r_{m-v}, \\ q_{v+2m}(t) &= t r_{m-1}. \end{aligned}$$

Moreover

$$r_m + 2t r_{m-1} + t^2 r_{m-2} - r_{m-v} = 1 \qquad (m \geq 0).$$

Proof. Rationalizing with respect to the odd power x^v gives

$$\begin{aligned} Q_{t,v}(x) &= \frac{(1+x^v)(1+tx^2 - x^v)}{(1-x^2)((1+tx^2)^2 - x^{2v})} \\ &= \frac{1 + tz - z^v + tx^v z}{(1-z)((1+tz)^2 - z^v)}. \end{aligned}$$

Separating the even part and the x^v times an even series gives the two coefficient identities. The recurrence for r_m is the coefficient of z^m in $((1+tz)^2 - z^v)R_{t,v}(z) = (1-z)^{-1}$. □

For $k, m \geq 0$ put

$$P_{k,m}(t) := \sum_{\ell=0}^{m} (-1)^\ell \binom{k+\ell}{\ell} t^\ell, \qquad P_{k,\infty}(t) = (1+t)^{-k-1}.$$

These are the Taylor partial sums of $(1+t)^{-k-1}$. The next theorem turns the parity rationalization into an infinite forbidden region for the two-predecessor zero pair.

Theorem 7.8 (Predecessor safe lattice)**.** *Let $0 < t < 1/2$, let $v \geq 5$ be odd, and let $n > 2v$. If*

$$q_{n-2}(t) = q_{n-v}(t) = 0,$$

then

$$n \geq 5v + 4 \quad \textit{when } n \textit{ is odd}, \qquad n \geq 6v + 2 \quad \textit{when } n \textit{ is even}.$$

Thus a first coefficient-one escape in Proposition 7.6 obeys the same lower bounds.

Proof. Write $E_m = q_{2m}$. Lemma 7.7 and its recurrence give

$$E_m = 1 - t r_{m-1} - t^2 r_{m-2}. \tag{7.2}$$

If n is odd, put $j = (n-v)/2 - 2$; if n is even, put $j = n/2 - v - 1$. The predecessor pairs become, respectively,

$$(q_{n-2}, q_{n-v}) = (tr_j, E_{j+2}), \qquad (q_{n-v}, q_{n-2}) = (tr_j, E_{j+v}). \tag{7.3}$$

It is therefore enough to exclude a common zero in either pair for $0 \le j < 2v$.

Expanding the defining rational function for $R_{t,v}$ geometrically in $z^v/(1+tz)^2$ gives, for $0 \le j < 2v$,

$$r_j = P_{1,j} + \mathbf{1}_{j \ge v} P_{3,j-v}. \tag{7.4}$$

The terms $(\ell+1)t^\ell$ decrease strictly when $0 < t < 1/2$, so every $P_{1,j}$ is positive. Taylor's theorem also gives $P_{k,2h} > P_{k,\infty} > 0$. Hence $r_j > 0$ unless $j = v + s$ with s odd. Odd $P_{3,s}$ increase with s, while for $s \ge 7$

$$r_{v+s} > (1+t)^{-2} + P_{3,7}(t) > 0.$$

For completeness, the last function is decreasing on $(0, 1/2)$: after differentiation its negative numerator is twice

$$420t^9 + 1008t^8 + 644t^7 + 14t^6 + (10t^4 - 12t^5) + (6t^2 - 8t^3) + (3 - 4t),$$

which is positive term by term. Its value at $t = 1/2$ is $37/144$. Consequently only $s \in \{1, 3, 5\}$ can remain.

First consider the odd pair in (7.3). At a zero of r_{v+s}, equation (7.2) gives

$$E_{v+s+2} = 1 - tr_{v+s+1}.$$

Here $P_{1,v+s+1} < (1+t)^{-2}$, and the even partial sums $P_{3,2h}$ decrease from $P_{3,2}$ for $h \ge 1$. Thus

$$r_{v+s+1} < (1+t)^{-2} + P_{3,2}(t).$$

The resulting lower bound

$$1 - t\big((1+t)^{-2} + 1 - 4t + 10t^2\big)$$

is positive on $(0, 1/2)$. Indeed, after multiplication by $(1+t)^2$ its complement is $10t^5 + 16t^4 + 3t^3 - 3t^2$, which is at most $15/16$: use $10t^5 \le 5t^4$, $t^4 \le t^2/4$, and $t^3 \le t^2/2$. Therefore the odd pair never vanishes for $j < 2v$.

It remains to treat the even pair for the three exceptional offsets. We will use the exact identity

$$E_{2v+s} = P_{0,2v+s} - t\big(P_{4,s-1} + P_{2,v+s-1}\big), \qquad s \in \{1, 3, 5\}, \tag{7.5}$$

obtained from the same geometric expansion. The odd partial sums $P_{2,m}$ increase for odd $m \ge 5$ and are positive: $P_{2,5}$ is decreasing on $(0, 1/2)$ because

$$P'_{2,5} = -3\{t^2(35t^2 - 20t + 3) + (7t^2 - 4t + 1)\} < 0,$$

and $P_{2,5}(1/2) = 1/32$.

For $s = 1$, a zero of r_{v+1} satisfies $P_{1,v+1} = 4t - 1$. The even $P_{1,2h}$ decrease, so $P_{1,6} - (4t-1) \ge 0$. This difference decreases on $(0, 1/2)$ and at $t = 383/1000$ equals

$$-\frac{2424992339692617}{10^{18}} < 0.$$

Hence $t < 383/1000$. Since $P_{0,2v+1} > 1 - t$ and $P_{2,v} < (1+t)^{-3}$, equation (7.5) gives

$$E_{2v+1} > 1 - 2t - \frac{t}{(1+t)^3} > 0;$$

the last expression decreases and at $t = 383/1000$ is already

$$\frac{117994119779}{1322624443500} > 0.$$

For $s = 3$, the function $(1+t)^{-2} + P_{3,3}(t)$ is decreasing. At $t = 433/1000$ its value is

$$\frac{637744741107}{102674450000000} > 0.$$

Thus a zero of r_{v+3} has $t > 433/1000$. Since $P_{0,2v+3} < (1+t)^{-1}$ and the P_2 term in (7.5) is positive,

$$E_{2v+3} < (1+t)^{-1} - tP_{4,2}(t) < 0.$$

The right side decreases beyond 433/1000 and at that endpoint is

$$-\frac{4430649363}{286600000000}.$$

For $s = 5$, direct differentiation shows that $(1+t)^{-2} + P_{3,5}(t)$ is decreasing; its derivative has negative numerator twice

$$140t^7 + 350t^6 + 240t^5 + 10t^4 + (6t^2 - 8t^3) + (3 - 4t) > 0.$$

At $t = 487/1000$ its value is the positive number

$$\frac{83847268664253747}{138198062500000000000},$$

so a zero of r_{v+5} has $t > 487/1000$. Equation (7.5) then gives

$$E_{2v+5} < (1+t)^{-1} - tP_{4,4}(t) < 0;$$

the endpoint value is $-46095940700109163/148700000000000000$, and the bound decreases thereafter. This excludes the even pair for every $j < 2v$.

Finally, $j < 2v$ corresponds to $n \le 5v + 2$ in the odd case and to $n \le 6v$ in the even case. Advancing by the relevant parity proves the two stated lower bounds. □

The accompanying dependency-free checker verifies 11544 two-block formulas, 17390 parity identities, and 23088 predecessor-pair identities for all 74 odd exponents through $v = 151$. An independent exact symbolic checker clears denominators and uses rational real-root counts to certify the eight derivative signs and two positivity assertions used above. These finite audits support, but are not logically substituted for, the uniform analytic proof.

Theorem 7.9 (Three-block predecessor safe lattice)**.** *Let $0 < t < 1/2$, let $v \ge 5$ be odd, and let $n > 2v$. If*

$$q_{n-2}(t) = q_{n-v}(t) = 0,$$

then

$$n \ge 7v + 4 \quad \textit{when } n \textit{ is odd}, \qquad n \ge 8v + 2 \quad \textit{when } n \textit{ is even}.$$

Thus every first coefficient-one escape in Proposition 7.6 obeys these stronger bounds.

Proof. Theorem 7.8 and (7.3) reduce the new work to $j = 2v + s$ with $0 \le s < v$. A third term in the geometric expansion used for (7.4) gives

$$r_{2v+s} = P_{1,2v+s} + P_{3,v+s} + P_{5,s}. \tag{7.6}$$

We use repeatedly the derivative identity $P'_{k,m} = -(k+1)P_{k+1,m-1}$ and the paired-increment identity

$$P_{k,m+2} - P_{k,m} = (-1)^{m+1}\binom{k+m+1}{m+1}t^{m+1}\left(1 - \frac{k+m+2}{m+2}t\right). \tag{7.7}$$

Thus the relevant odd partial sums increase with their odd order and the relevant even partial sums decrease with their even order.

We first isolate all possible zeros of (7.6). If s is even and $s \geq 2$, or if $s = 0$ and $v \geq 7$, Taylor's theorem and (7.7) give

$$r_{2v+s} > (1+t)^{-2} + P_{3,7}(t) + (1+t)^{-6} > 0.$$

The last function is decreasing, and its value at $1/2$ is $4021/11664$. For the remaining case $(v, s) = (5, 0)$, use $P_{3,5}(t) \geq P_{3,5}(1/2) = -9/16$ and $P_{1,10} > (1+t)^{-2} \geq 4/9$ to obtain $r_{10} > 127/144$. If $s \geq 13$ is odd, then $v \geq 15$ and

$$r_{2v+s} > P_{1,43}(t) + (1+t)^{-4} + P_{5,13}(t) > 0.$$

Again the displayed function is decreasing; at $1/2$ it equals

$$\frac{108675557490085}{356241767399424} > 0.$$

Hence only $s \in \{1, 3, 5, 7, 9, 11\}$ can remain.

For these six offsets let v_s be the least admissible odd v, namely $5, 5, 7, 9, 11, 13$, and put

$$L_s = P_{1,2v_s+s} + (1+t)^{-4} + P_{5,s}, \qquad U_s = (1+t)^{-2} + P_{3,v_s+s} + P_{5,s}.$$

Odd partial sums increase and even partial sums decrease, so $L_s < r_{2v+s} < U_s$. Exact rational Sturm chains give the following brackets for every zero of r_{2v+s} in $(0, 1/2)$:

s	1	3	5	7	9	11
α_s	.3178	.3441	.3873	.4270	.4618	.4921
β_s	.322	.345	.388	.428	.462	.493

$\alpha_s < t < \beta_s.$

All decimals here are the displayed terminating rationals. The certificate checks that L_s has no zero on $(0, \alpha_s]$ and is positive there, while U_s has no zero on $[\beta_s, 1/2)$ and is negative there. Their endpoint margins are retained as exact fractions.

Consider first the odd event pair. At a zero of r_{2v+s}, taking one more partial-sum term in (7.6) gives

$$r_{2v+s+1} = (2v+s+2)t^{2v+s+1} - \binom{v+s+4}{3}t^{v+s+1} + \binom{s+6}{5}t^{s+1}.$$

Dropping the negative middle term and using $t < \beta_s$, the remaining v-dependent term is maximal at $v = v_s$: after replacing v by $v+2$, its ratio is at most

$$\frac{2v+s+6}{2v+s+2}\beta_s^4 \leq \frac{17}{13}\left(\frac{493}{1000}\right)^4 < 1.$$

We therefore obtain

$$tr_{2v+s+1} < \binom{s+6}{5}\beta_s^{s+2} + (2v_s+s+2)\beta_s^{2v_s+s+2} < \frac{3}{4}.$$

The last six strict rational inequalities are recorded in the audit. Thus $E_{2v+s+2} = 1 - tr_{2v+s+1} > 1/4$, excluding the odd pair.

For the even pair, the same geometric expansion gives, for every m,

$$E_m = P_{0,m} - t \sum_{1 \le h \le m/v} P_{2h,m-hv-1}. \tag{7.8}$$

At $m = 3v + s$ this has the three terms P_2, P_4, P_6 in the sum. If $s = 1$, the root bracket and partial-sum monotonicity imply

$$E_{3v+1} > (1+t)^{-1} - t\{P_{2,10}(t) + (1+t)^{-5} + 1\} > 0.$$

The final function has no zero on $[.3178, .322]$ and is positive at both endpoints. For $s \in \{3, 5, 7, 9, 11\}$, use $P_{0,3v+s} = (1 + t^{3v+s+1})/(1+t)$, bound the positive tail by $\beta_s^{3v_s+s+1}$, and use monotonicity in the other three partial sums:

$$\begin{aligned} E_{3v+s} < (1+t)^{-1} &+ \beta_s^{3v_s+s+1} \\ &- t\{(1+t)^{-3} + P_{4,v_s+s-1}(t) + P_{6,s-1}(t)\} < 0. \end{aligned}$$

For each s, the last upper bound has no zero on $[\alpha_s, \beta_s]$ and is negative at both endpoints, again by exact rational Sturm chains. This excludes the even pair.

We have excluded both pairs for $j < 3v$. The largest corresponding odd event index is $7v + 2$, and the largest even event index is $8v$. Advancing by the relevant parity proves the theorem. □

The dependency-free checker verifies 30591 three-block formulas, 40887 parity identities, and 61182 predecessor-pair identities for all 99 odd exponents through $v = 201$. A separate symbolic checker supplies 21 exact root-free rational-interval sign certificates for the exceptional offsets and companion bounds. These audits are reproducible finite checks of the explicit algebra in the uniform proof; the theorem is not inferred from the cutoff $v = 201$.

Theorem 7.10 (Four-block predecessor safe lattice)**.** *Let $0 < t < 1/2$, let $v \ge 5$ be odd, and let $n > 2v$. If*

$$q_{n-2}(t) = q_{n-v}(t) = 0,$$

then

$$n \ge 9v + 4 \quad \textit{when } n \textit{ is odd}, \qquad n \ge 10v + 2 \quad \textit{when } n \textit{ is even}.$$

Thus every first coefficient-one escape in Proposition 7.6 obeys these stronger bounds.

Proof. By Theorem 7.9, it remains to exclude the two pairs in (7.3) for $j = 3v+s$, $0 \le s < v$. The next geometric term gives

$$r_{3v+s} = P_{1,3v+s} + P_{3,2v+s} + P_{5,v+s} + P_{7,s}. \tag{7.9}$$

We continue to use the paired-increment identity (7.7).

First isolate the zeros of (7.9). Suppose that s is even. For $s \in \{2, 4, \dots, 16\}$, let $w_s = \max(5, s+1)$. Odd partial sums increase and even partial sums decrease toward their limits, so

$$r_{3v+s} > P_{1,3w_s+s} + (1+t)^{-4} + P_{5,w_s+s} + P_{7,s} > 0.$$

The last eight signs are exact Sturm certificates. The same argument with $w_0 = 9$ proves positivity for $s = 0$ and $v \ge 9$. If $s \ge 18$ is even, then $P_{1,3v+s} > 0$, while the P_3 and P_7 terms are positive even partial sums. Also $v + s \ge 2s + 1 \ge 37$, and $P_{5,v+s} \ge P_{5,37} > 0$; the last sign is again certified on $(0, 1/2)$. Thus the only even-offset cases not covered analytically are $(v, s) = (5, 0)$ and $(7, 0)$.

Now suppose that s is odd. For $s \geq 19$, the least admissible exponent is $v = s + 2$, and monotonicity gives

$$r_{3v+s} > (1+t)^{-2} + P_{3,3s+4} + (1+t)^{-6} + P_{7,s} \geq (1+t)^{-2} + P_{3,61} + (1+t)^{-6} + P_{7,19} > 0.$$

The final function has no zero on $(0, 1/2)$ and is positive at both endpoints. For $s \in \{1, 3, \ldots, 17\}$, put $v_1 = 7$ and $v_s = \max(5, s+2)$ otherwise, and define

$$L_s = (1+t)^{-2} + P_{3,2v_s+s} + (1+t)^{-6} + P_{7,s},$$
$$U_s = P_{1,3v_s+s} + (1+t)^{-4} + P_{5,v_s+s} + P_{7,s}.$$

Then $L_s < r_{3v+s} < U_s$. Exact rational Sturm chains confine every zero to the following intervals:

s	1	3	5	7	9
α_s	.277	.288	.323	.359	.391
β_s	.279	.290	.325	.360	.392

s	11	13	15	17
α_s	.420	.446	.469	.490
β_s	.421	.447	.471	.492

Here L_s is positive and root-free on $(0, \alpha_s]$, while U_s is negative and root-free on $[\beta_s, 1/2)$. The only omitted odd case is $(v, s) = (5, 1)$.

The three finite cases $(5, 0)$, $(7, 0)$, and $(5, 1)$ are disposed of without approximating their roots. In each case the two integer-polynomial pairs (r_{3v+s}, E_{3v+s+2}) and (r_{3v+s}, E_{4v+s}) are primitive, their degrees are preserved modulo 7, and their monic gcds in $\mathbb{F}_7[t]$ are 1. The retained certificates give every Euclidean remainder. Gauss's lemma therefore excludes a common zero over $\mathbb{C}$, and hence over $(0, 1/2)$.

It remains to separate the companions at the nine uniform root intervals. At a zero of r_{3v+s}, the next partial-sum terms give

$$\begin{aligned} r_{3v+s+1} = & -(3v+s+2)t^{3v+s+1} + \binom{2v+s+4}{3} t^{2v+s+1} \\ & - \binom{v+s+6}{5} t^{v+s+1} + \binom{s+8}{7} t^{s+1}. \end{aligned}$$

After multiplication by t, discard the two negative terms. The ratio of the remaining v-dependent term after replacing v by $v+2$ is at most

$$\frac{\binom{2v+s+8}{3}}{\binom{2v+s+4}{3}} \beta_s^4 < 1;$$

it is enough to check this at $v = v_s$. The nine exact rational margins give

$$t r_{3v+s+1} < \binom{2v_s+s+4}{3} \beta_s^{2v_s+s+2} + \binom{s+8}{7} \beta_s^{s+2} < 1.$$

Consequently $E_{3v+s+2} = 1 - t r_{3v+s+1} > 0$, excluding the odd event pair.

For the even event pair, substitute $m = 4v + s$ in (7.8). When $s = 1$, partial-sum monotonicity and the root bracket give

$$\begin{aligned} E_{4v+1} > & \frac{1 - \beta_1^{4v_1+2}}{1+t} \\ & - t\{(1+t)^{-3} + P_{4,2v_1}(t) + (1+t)^{-7} + 1\} > 0. \end{aligned}$$

For $s \in \{3, 5, \ldots, 17\}$, the complementary bounds give

$$E_{4v+s} < (1+t)^{-1} - t\{P_{2,3v_s+s-1}(t) + (1+t)^{-5} + P_{6,v_s+s-1}(t) + P_{8,s-1}(t)\} < 0.$$

Each displayed companion bound is root-free with the stated sign on $[\alpha_s, \beta_s]$, by exact Sturm chains. This excludes the even event pair.

We have excluded both pairs for $j < 4v$. The largest corresponding odd event index is $9v + 2$, and the largest even event index is $10v$. Advancing by the relevant parity proves the theorem. □

The dependency-free checker verifies 63488 four-block formulas, 79484 parity identities, and 126976 predecessor-pair identities for all 124 odd exponents through $v = 251$. It also retains full mod-7 Euclidean certificates for the six finite exceptional companion pairs. An independent SymPy checker supplies 38 root-free rational-interval sign certificates and recomputes all six modular gcds. Both deterministic JSON outputs use canonical LF bytes, making their recorded hashes platform-independent. The finite cutoff audits the algebra; the theorem itself is uniform.

Theorem 7.11 (Five-block predecessor safe lattice)**.** *Let $0 < t < 1/2$, let $v \geq 5$ be odd, and let $n > 2v$. If*

$$q_{n-2}(t) = q_{n-v}(t) = 0,$$

then

$$n \geq 11v + 4 \quad \text{when } n \text{ is odd}, \qquad n \geq 12v + 2 \quad \text{when } n \text{ is even}.$$

Thus every first coefficient-one escape in Proposition 7.6 obeys these stronger bounds.

Proof. By Theorem 7.10, it remains to exclude the two pairs in (7.3) for $j = 4v+s$, $0 \leq s < v$. The next geometric term gives

$$r_{4v+s} = P_{1,4v+s} + P_{3,3v+s} + P_{5,2v+s} + P_{7,v+s} + P_{9,s}. \tag{7.10}$$

We first isolate its zeros. Suppose that s is even. For $s \in \{0, 2, 4, 6\}$ use respectively

$$(s, v_0) = (0, 17), (2, 7), (4, 5), (6, 7)$$

and put

$$M_{s,v_0} = (1 + t)^{-2} + P_{3,3v_0+s} + (1 + t)^{-6} + P_{7,v_0+s} + P_{9,s}.$$

Even partial sums lie above their limits and odd partial sums increase with their odd order. Hence $r_{4v+s} > M_{s,v_0}$ whenever $v \geq v_0$. Exact rational Sturm chains show that these four functions are positive and root-free on $(0, 1/2)$. This leaves only

$$(v, s) = (5, 0), (5, 2), (7, 0), (9, 0), (11, 0), (13, 0), (15, 0). \tag{7.11}$$

For every even $s \geq 8$, the same monotonicity gives

$$r_{4v+s} > M_s := (1 + t)^{-2} + P_{3,4s+3} + (1 + t)^{-6} + P_{7,2s+1} + P_{9,s}.$$

Write $A_9 = \binom{s+10}{9}$, $A_7 = \binom{2s+9}{7}$, and $A_3 = \binom{4s+7}{3}$. Alternating-tail estimates give

$$\begin{aligned} P_{9,s} - (1 + t)^{-10} &\geq A_9 t^{s+1} \left(1 - \frac{s + 11}{s + 2} t\right), \\ (1 + t)^{-8} - P_{7,2s+1} &< A_7 t^{2s+2}, \\ (1 + t)^{-4} - P_{3,4s+3} &< A_3 t^{4s+4}. \end{aligned}$$

It is enough that $R_7(s) + R_3(s) < 1$, where

$$R_7(s) = \frac{A_7}{A_9} \frac{2(s + 2)}{s - 7} 2^{-s-1}, \qquad R_3(s) = \frac{A_3}{A_9} \frac{2(s + 2)}{s - 7} 2^{-3s-3}.$$

At $s = 8$ their sum is $126012629939/326283296768 < 1$. Both summands decrease under $s \mapsto s+2$: their successive ratios are

$$\frac{(s-7)(s+4)(s+5)(s+6)(2s+11)(2s+13)}{4(s-5)(s+2)(s+11)(s+12)(2s+3)(2s+5)},$$
$$\frac{(s-7)(s+3)(s+4)(2s+7)(4s+13)(4s+15)}{64(s-5)(s+11)(s+12)(2s+3)(4s+5)(4s+7)}.$$

After $s = x+8$, the numerators of one minus these ratios have only positive coefficients. Thus $M_s > 0$, excluding every remaining even offset.

Now let s be odd. Put $v_1 = 13$, and for $s \in \{3, 5, \ldots, 23\}$ put $v_s = \max(5, s+2)$. Define

$$L_s = P_{1,4v_s+s} + (1+t)^{-4} + P_{5,2v_s+s} + (1+t)^{-8} + P_{9,s},$$
$$U_s = (1+t)^{-2} + P_{3,3v_s+s} + (1+t)^{-6} + P_{7,v_s+s} + P_{9,s}.$$

For $v \geq v_s$, the paired-increment identity gives $L_s < r_{4v+s} < U_s$. For odd $s \geq 25$, the lower bound increases with s, and its first instance

$$P_{1,133} + (1+t)^{-4} + P_{5,79} + (1+t)^{-8} + P_{9,25}$$

is positive and root-free on $(0, 1/2)$ by an exact Sturm chain. Hence only the twelve offsets $s = 1, 3, \ldots, 23$ remain. Exact rational Sturm chains confine their uniform zeros to

s	1	3	5	7	9	11
α_s	.248	.249	.279	.310	.340	.367
β_s	.250	.251	.280	.312	.341	.368

s	13	15	17	19	21	23
α_s	.392	.415	.435	.454	.472	.488
β_s	.393	.416	.437	.456	.474	.490.

Here L_s is positive and root-free on $(0, \alpha_s]$, while U_s is negative and root-free on $[\beta_s, 1/2)$. At $s = 1$ this uses $v \geq 13$; the omitted cases are

$$(v, s) = (5, 1), (7, 1), (9, 1), (11, 1). \tag{7.12}$$

All eleven cases in (7.11) and (7.12) are disposed of without approximating a root. For each case both integer-polynomial pairs

$$(r_{4v+s}, E_{4v+s+2}), \qquad (r_{4v+s}, E_{5v+s})$$

are primitive and have degree-preserving reduction with monic gcd 1 over a finite field. Modulus 7 works except for $(v, s, \text{companion}) = (5, 0, \text{odd}), (5, 0, \text{even}), (5, 2, \text{even})$, for which modulus 11 works. The retained certificates contain every Euclidean remainder. Gauss's lemma excludes a common zero over $\mathbb{C}$.

It remains to separate the companions on the twelve uniform intervals. At a zero of r_{4v+s}, the five next partial-sum terms give

$$r_{4v+s+1} = (4v+s+2)t^{4v+s+1} - \binom{3v+s+4}{3} t^{3v+s+1} + \binom{2v+s+6}{5} t^{2v+s+1} - \binom{v+s+8}{7} t^{v+s+1} + \binom{s+10}{9} t^{s+1}.$$

After multiplication by t, discard the two negative terms. The two v-dependent positive terms decrease under $v \mapsto v+2$ on the root interval because

$$\frac{4(v_s+2)+s+2}{4v_s+s+2}\,\beta_s^8 < 1, \qquad \frac{\binom{2(v_s+2)+s+6}{5}}{\binom{2v_s+s+6}{5}}\,\beta_s^4 < 1.$$

The twelve exact margins give

$$(4v_s+s+2)\beta_s^{4v_s+s+2}+\binom{2v_s+s+6}{5}\beta_s^{2v_s+s+2}+\binom{s+10}{9}\beta_s^{s+2}<1.$$

Consequently $E_{4v+s+2}=1-tr_{4v+s+1}>0$.

For the even event pair, substitute $m=5v+s$ in (7.8). At $s=1$, partial-sum monotonicity gives

$$E_{5v+1}>(1+t)^{-1}-t\{P_{2,4v_1}+(1+t)^{-5}+P_{6,2v_1}+(1+t)^{-9}+1\}>0.$$

For $s\in\{3,5,\dots,23\}$, complementary bounds give

$$\begin{aligned}E_{5v+s}<{}&\frac{1+t^{5v_s+s+1}}{1+t}\\&-t\{(1+t)^{-3}+P_{4,3v_s+s-1}+(1+t)^{-7}+P_{8,v_s+s-1}+P_{10,s-1}\}<0.\end{aligned}$$

Each displayed bound is root-free with the stated sign on $[\alpha_s,\beta_s]$, by exact rational Sturm chains.

We have excluded both pairs for $j<5v$. The largest corresponding odd event index is $11v+2$, and the largest even event index is $12v$. Advancing by the relevant parity proves the theorem. □

The dependency-free checker verifies 79360 five-block formulas, 95356 parity identities, and 158720 predecessor-pair identities for all 124 odd exponents through $v=251$, together with full modular Euclidean certificates for the 22 finite exceptional pairs. An independent SymPy checker supplies 41 root-free rational-interval sign certificates and recomputes all 22 modular gcds. A separate exact reconnaissance through $v=47$ records the discovery pattern but is not used to infer the theorem. All deterministic outputs use canonical LF bytes; the finite audit windows check the algebra, while the proof above is uniform.

Theorem 7.12 (Six-block predecessor safe lattice)**.** *Let $0<t<1/2$, let $v\geq 5$ be odd, and let $n>2v$. If*

$$q_{n-2}(t)=q_{n-v}(t)=0,$$

then

$$n\geq 13v+4\quad\textit{when } n \textit{ is odd},\qquad n\geq 14v+2\quad\textit{when } n \textit{ is even}.$$

Thus every first coefficient-one escape in Proposition 7.6 obeys these stronger bounds.

Proof. By Theorem 7.11, it remains to exclude (7.3) for $j=5v+s$, $0\leq s<v$. The sixth geometric term gives

$$r_{5v+s}=P_{1,5v+s}+P_{3,4v+s}+P_{5,3v+s}+P_{7,2v+s}+P_{9,v+s}+P_{11,s}. \tag{7.13}$$

We first exclude even s. For

$$(s,v_0)=(0,23),(2,15),(4,5),(6,7),(8,9)$$

put

$$M_{s,v_0}=P_{1,5v_0+s}+(1+t)^{-4}+P_{5,3v_0+s}+(1+t)^{-8}+P_{9,v_0+s}+P_{11,s}.$$

Odd partial sums increase with their odd order and even partial sums lie above their limits. Hence $r_{5v+s}>M_{s,v_0}$ for $v\geq v_0$. Exact rational Horner interval enclosures certify $M_{s,v_0}>0$ on $(0,1/2)$ in all five cases. The omitted pairs are

$$\{(v,0):5\leq v\leq 21\}\cup\{(v,2):5\leq v\leq 13\},\qquad v \text{ odd}. \tag{7.14}$$

For every even $s \geq 10$, take $v_0 = s + 1$. The same lower bound applies. Write

$$A_{11} = \binom{s+12}{11}, \quad A_9 = \binom{2s+11}{9}, \quad A_5 = \binom{4s+9}{5}, \quad A_1 = 6s + 7.$$

The alternating remainder bounds compare the positive remainder of $P_{11,s}$ with the three negative remainders:

$$\begin{aligned} P_{11,s} - (1+t)^{-12} &\geq A_{11} t^{s+1} \left(1 - \frac{s+13}{s+2} t\right), \\ (1+t)^{-10} - P_{9,2s+1} &< A_9 t^{2s+2}, \\ (1+t)^{-6} - P_{5,4s+3} &< A_5 t^{4s+4}, \\ (1+t)^{-2} - P_{1,6s+5} &< A_1 t^{6s+6}. \end{aligned}$$

It is enough that

$$\frac{2(s+2)}{s-9} \frac{1}{A_{11}} \left(\frac{A_9}{2^{s+1}} + \frac{A_5}{2^{3s+3}} + \frac{A_1}{2^{5s+5}} \right) < 1.$$

At $s = 10$ the left side is $27281522928263929231/81461110059877531648 < 1$. Each of its three summands decreases under $s \mapsto s + 2$; after $s = x + 10$, the exact numerators of one minus the successive ratios have only positive coefficients. This excludes every even offset outside (7.14).

Now let s be odd. Put $v_1 = 19$, $v_3 = 11$, and $v_s = s + 2$ for $s \geq 5$. Define

$$\begin{aligned} L_s &= (1+t)^{-2} + P_{3,4v_s+s} + (1+t)^{-6} + P_{7,2v_s+s} + (1+t)^{-10} + P_{11,s}, \\ U_s &= P_{1,5v_s+s} + (1+t)^{-4} + P_{5,3v_s+s} + (1+t)^{-8} + P_{9,v_s+s} + P_{11,s}. \end{aligned}$$

Paired increments give $L_s < r_{5v+s} < U_s$ for $v \geq v_s$. For odd $s \geq 33$, the three negative Taylor remainders in L_s are bounded at $t = 1/2$ by

$$\binom{s+12}{11} 2^{-s-1} + \binom{3s+12}{7} 2^{-3s-5} + \binom{5s+12}{3} 2^{-5s-9}.$$

At $s = 33$ this is

$$\frac{1768483226579352197428635688881857408474171792538535}{29931553532536891764811465374029476242553498480148 48} < \frac{421876}{531441} = \sum_{k=0}^{5} \left(\frac{2}{3}\right)^{2k+2}.$$

All three errors decrease under $s \mapsto s + 2$, again by positive-coefficient ratio certificates. Hence $L_s > 0$ for every odd $s \geq 33$.

The remaining offsets $s = 1, 3, \ldots, 31$ are confined to the following rational intervals:

s	1	3	5	7	9	11	13	15
α_s	.226	.220	.246	.274	.301	.326	.350	.371
β_s	.228	.222	.248	.276	.303	.329	.352	.374

s	17	19	21	23	25	27	29	31
α_s	.391	.410	.427	.443	.458	.472	.486	.499
β_s	.394	.413	.430	.446	.462	.476	.489	.4999

Exact rational interval Horner certificates show $L_s > 0$ on $(0, \alpha_s]$ and $U_s < 0$ on $[\beta_s, 1/2)$. The cases omitted by the choices of v_1 and v_3 are

$$\{(v, 1) : 5 \leq v \leq 17\} \cup \{(v, 3) : 5 \leq v \leq 9\}, \qquad v \text{ odd}. \tag{7.15}$$

Every one of the 24 cases in (7.14)–(7.15) is disposed of algebraically. For both pairs

$$(r_{5v+s}, E_{5v+s+2}), \qquad (r_{5v+s}, E_{6v+s}),$$

the integer polynomials are primitive and have degree-preserving reduction with monic gcd 1 over one of $\mathbb{F}_2, \mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$. The retained 48 certificates contain every Euclidean remainder; Gauss's lemma excludes a common complex zero.

It remains to separate the companions on the 16 uniform intervals. At a zero of r_{5v+s}, the six next terms alternate. Discarding the three negative terms gives

$$tr_{5v+s+1} < \binom{4v+s+4}{3} t^{4v+s+2} + \binom{2v+s+8}{7} t^{2v+s+2} + \binom{s+12}{11} t^{s+2}.$$

The two v-dependent terms decrease under $v \mapsto v+2$ on each root interval, and their exact endpoint sum is strictly below 1 at $v = v_s$. Thus $E_{5v+s+2} = 1 - tr_{5v+s+1} > 0$.

Finally, substitute $m = 6v + s$ in (7.8). For $s = 1$, partial-sum monotonicity gives the positive lower bound

$$E_{6v+1} > \frac{1 - t^{6v_1+2}}{1+t} - t\{(1+t)^{-3} + P_{4,4v_1} + (1+t)^{-7} + P_{8,2v_1} + (1+t)^{-11} + 1\} > 0.$$

For $s = 3, 5, \ldots, 31$, complementary bounds give

$$E_{6v+s} < \frac{1}{1+t} - t\{P_{2,5v_s+s-1} + (1+t)^{-5} + P_{6,3v_s+s-1} + (1+t)^{-9} + P_{10,v_s+s-1} + P_{12,s-1}\} < 0.$$

All 16 signs hold throughout $[\alpha_s, \beta_s]$ by the retained exact interval certificates. Hence both predecessor pairs are excluded for $j < 6v$. The largest odd and even event indices are $13v+2$ and $14v$; advancing by parity proves the theorem. □

The dependency-free checker verifies 95232 six-block formulas, 111228 parity identities, and 190464 predecessor-pair identities for all 124 odd exponents through $v = 251$, together with full modular Euclidean certificates for the 48 finite pairs. An independent SymPy checker supplies 53 exact rational interval-Horner sign certificates and recomputes all 48 modular gcds. Both theorem-level JSON outputs replay byte-for-byte. Exact finite reconnaissance through $v = 23$ records 89 roots and constant companion gcds; it is discovery evidence only and is not used in the uniform proof.

Remark 7.13 (An exact open bottleneck). Regard the q_j as polynomials in an indeterminate t. Exact modular gcd certificates accompanying this paper prove

$$\gcd_{\mathbb{Z}[t]}(q_{n-2}(t), q_{n-v}(t)) = 1$$

for all 876456 pairs with odd $5 \le v \le 151$ and $2v < n \le 12000$, with no unresolved index. This is finite evidence, not an infinite proof. Theorem 7.12 independently removes every real event below its parity-dependent linear thresholds. Proving the displayed coprimality for every odd $v \ge 5$ and $n > 2v$ would eliminate the only branch identified by Proposition 7.6. Together with Lemma 7.3, it would settle the entire normalized smaller-gap-2 trinomial family, without restricting the algebraic degree of a.

We now use the partial sums above to give a finite exact representation of all coefficients needed in Lemma 7.5. Expanding (7.1) first in powers of x^v gives

$$Q_{a,v}(x) = \frac{1}{(1-x^2)(1+ax^2)} + \sum_{k\geq 1} \frac{(-1)^{k-1} a x^{kv+2}}{(1-x^2)(1+ax^2)^{k+1}}. \tag{7.16}$$

Since v is odd, equal parities combine. For $m \geq 0$ this yields

$$q_{Kv+2+2m} = a \sum_{s=0}^{(K-1)/2} P_{K-2s,m+sv}(a), \qquad K \text{ odd}, \tag{7.17}$$

$$q_{Kv+2+2m} = P_{0,(K/2)v+1+m}(a) - a \sum_{s=0}^{K/2-1} P_{K-2s,m+sv}(a), \qquad K \text{ even}. \tag{7.18}$$

These identities are formal and exact. Keeping the top partial sum fixed and sending odd v to infinity gives

$$L_{K,m}(a) = \begin{cases} aP_{K,m}(a) + a \displaystyle\sum_{\substack{1\leq j<K \\ j \text{ odd}}} (1+a)^{-j-1}, & K \text{ odd}, \\ (1+a)^{-1} - aP_{K,m}(a) - a \displaystyle\sum_{\substack{2\leq j<K \\ j \text{ even}}} (1+a)^{-j-1}, & K \text{ even}. \end{cases} \tag{7.19}$$

For the uniform estimates, if $0 < a \leq U < 1/2$ and

$$U \frac{k+m+2}{m+2} < 1,$$

then the alternating terms after $P_{k,m}$ decrease in magnitude, and

$$\left| P_{k,m}(a) - (1+a)^{-k-1} \right| \leq \binom{k+m+1}{m+1} U^{m+1}. \tag{7.20}$$

Thus every block has an exact $v \to \infty$ limit, and all terms omitted from that limit have explicit rational bounds. Increasing odd v only decreases those bounds once (7.20) applies.

Theorem 7.14 (Global cubic exclusion)**.** *Let $C = A_a B$ be a normalized nonnegative factorization of a Newman polynomial, where*

$$A_a(x) = 1 + ax^2 + x^v, \qquad v > 2, \qquad a \notin \mathbb{Z}.$$

Then $[\mathbb{Q}(a) : \mathbb{Q}] \neq 3$. Consequently, in the smaller-gap-2 case every genuinely fractional coefficient has algebraic degree at least 4, without any restriction on the product degree.

Proof. Theorem 3.2 removes even v and the boundary cases; after reversal the remaining cases have odd $v \geq 5$. Lemma 7.2 reduces a cubic coefficient to 126 exact minimal-polynomial records.

For $v = 5, 7, \ldots, 21$, the verifier works directly in $\mathbb{Z}[a]/(m_a)$. Multiplication by a and the recurrence

$$q_n = 1 - aq_{n-2} - q_{n-v}$$

produce a triple of integers for every coefficient. Rational bisection of the unique target root in $(0, 1/2)$ separates every sign. In all 1134 catalog–case pairs, the only zero before the first negative coefficient is q_v (besides the initial odd zero ray). The verifier also certifies

the forcing term $aq_{n-2} + q_{n-v}$ strictly positive at every step, so each preceding coefficient is strictly less than 1.

For every odd $v \geq 23$, formulas (7.17) and (7.18) give a uniform certificate. Each catalog entry records one obstruction index

$$N = Kv + r, \qquad K \leq 27,$$
$$(K, r) \in \{(K, 4) : K \text{ odd}\} \cup \{(K, 6) : K \text{ even}\} \cup \{(K, 8) : K \text{ odd}\}.$$

For a fixed parity, the ranges $0 \leq m \leq v-1$ in consecutive values of K partition all coefficients after the initial rays. Proving a complete block positive for every $m \geq 0$ is therefore stronger than the required statement. The base block $K = 0$ is elementary:

$$q_{2+2m} = P_{0,m+1}(a) = \frac{1 - (-a)^{m+2}}{1 + a} \in (0, 1) \qquad (0 \leq m \leq v - 1,\ 0 < a < 1).$$

For every block $K \geq 1$, using an outward dyadic interval containing the target root, the verifier checks every internal block offset $0 \leq m \leq 40$ by rational interval arithmetic. It then uses (7.20) to prove all later offsets positive. All complete blocks before K are positive, as is the prefix of block K before r; the only preceding noninitial zero is $q_v = 0$. Finally, the same tail estimate proves $q_{Kv+r} < 0$ uniformly from $v = 23$ onward. The weakest retained rational upper margin is less than $-2.29 \cdot 10^{-4}$, while the weakest pre-obstruction lower margin is greater than $5.19 \cdot 10^{-4}$. These decimals merely summarize exact fractions.

An independent identity checker compares 29193 coefficients obtained by direct formal division with the block expansion at $v = 23$. A second run at $v = 99$ compares 123053 identities. A separate rational-specialization checker, independent of the cubic quotient arithmetic, compares 9009 more identities. A stress checker iterates the algebraic recurrence directly for all 4914 pairs with odd $23 \leq v \leq 99$, without using the block-limit inequalities. These audits are not needed for the infinite conclusion. Lemma 7.5 now supplies the contradiction for every catalog entry and every odd $v \geq 5$. Combining with Theorem 6.4 proves the asserted degree-at-least-four conclusion. □

8 Global quartic exclusion for the smaller gap two

The same forced-series mechanism extends one algebraic degree further, but the finite arithmetic layer is substantially larger. The catalog below is exhaustive under the conjugate bounds, and every retained sign is certified by integer or rational arithmetic.

Lemma 8.1 (Uniform quartic catalog)**.** *Suppose that $a \in (0, 1/2)$ is a quartic algebraic integer permitted by the norm and conjugate restrictions of the smaller-gap-2 case. Then its monic minimal polynomial*

$$m_a(T) = T^4 + s_3T^3 + s_2T^2 + s_1T + s_0$$

belongs to an explicit catalog of 6,977 *irreducible integer polynomials. Five catalog polynomials have two roots in $(0, 1/2)$ and all others have one, so the catalog contains* 6,982 *target roots.*

Proof. Every conjugate has modulus strictly less than $R = 143/50$, every real conjugate is greater than -2, and the distinguished root satisfies $0 < a < 1/2$. Vieta's formulas therefore give the integer bounds

$$|s_3| \leq 9, \qquad |s_2| \leq 28, \qquad |s_1| \leq 35, \qquad |s_0| \leq 11.$$

For example, $|s_3| < 1/2 + 3R$, and the other inequalities follow by separating the elementary symmetric terms that contain a from those that do not. The resulting box contains 1,691,646 tuples with $s_0 \neq 0$.

The verifier applies the strict Schur recursion to $50^4 m_a((143/50)T)$, leaving 63,546 polynomials whose roots all lie in $|z| < R$. Exact integer linear- and quadratic-factor tests leave 57,238 irreducibles. Rational Sturm chains retain 10,671 polynomials having a root in $(0, 1/2)$ and then remove every polynomial with a real root in $(-3, -2]$; the disk bound makes this the entire forbidden real range. The remaining counts are exactly those stated. The same Sturm chains isolate every target root in a rational interval of width at most 2^{-36}. □

Theorem 8.2 (Global quartic exclusion)**.** *Let $C = A_a B$ be a normalized nonnegative factorization of a Newman polynomial, where*

$$A_a(x) = 1 + ax^2 + x^v, \qquad v > 2, \qquad a \notin \mathbb{Z}.$$

Then $[\mathbb{Q}(a) : \mathbb{Q}] \neq 4$. Consequently, in the smaller-gap-2 case every genuinely fractional coefficient has algebraic degree at least 5, without any restriction on the product degree.

Proof. As in Theorem 7.14, it is enough to consider odd $v \geq 5$ and the target roots in Lemma 8.1. For each of the 6,982 roots and every $v = 5, 7, \ldots, 21$, the verifier iterates

$$q_n = 1 - aq_{n-2} - q_{n-v}$$

exactly in $\mathbb{Z}[a]/(m_a)$. Rational interval evaluation separates every algebraic sign. Across all 62,838 root–exponent pairs, the only noninitial zero before the first negative coefficient is q_v, and every forcing term preceding that negative coefficient is strictly positive.

For odd $v \geq 23$, formulas (7.17)–(7.20) give a uniform certificate. The expanded obstruction search uses every parity-compatible even offset through 100: offsets divisible by 4 for odd blocks and offsets congruent to 2 modulo 4 for even blocks. Each target root has a certified obstruction

$$N = Kv + r, \qquad K \leq 110, \qquad r \leq 100,$$

whose exact limiting upper bound plus its finite-v error is negative. This expanded offset set is essential: restricting to offsets 4, 6, and 8 can skip an earlier negative coefficient at offset 10.

For every selected obstruction, a separate exact verifier proves all complete preceding blocks positive and also the prefix of the obstruction block before r. It evaluates internal offsets $0 \leq m \leq 48$ by rational interval arithmetic and uses the full-block limit with a first-omitted alternating-tail bound for every later offset. The base block is positive by the formula in the proof of Theorem 7.14. At every intervening index at least one of the two predecessor coefficients is therefore positive; hence $aq_{n-2} + q_{n-v} > 0$ and $q_n < 1$. Thus the hypotheses of Lemma 7.5 hold up to N, where the forced coefficient is negative. This contradicts the existence of the normalized nonnegative cofactor and excludes every quartic target root. □

The quartic proof objects retain the ordered minimal-polynomial catalog, both target intervals for the five double-target polynomials, every finite algebraic sign record, one exact uniform negative-tail certificate per target root, and the complete positive-prefix audit. Floating point is used only to order candidate obstruction checks; every accepted inequality is recomputed with exact rational endpoints.

For a final independent audit, a second implementation replaces the catalog generator's Schur recursion by the Schur–Cohn Hermitian determinant criterion, factors every survivor independently with FLINT, and encloses all complex roots with Arb. It reproduces the

ordered catalog and independently validates all 6,982 retained target intervals. At 4096-bit ball precision, a separate direct scalar recurrence checks all 62,838 finite root–exponent pairs, and an independent parity-block implementation checks every retained uniform tail and positive prefix certificate. In total these two sign audits certify 12,369,079 finite coefficient inequalities, 6,982 uniform negative-tail certificates, and 4,979,348 uniform prefix inequalities, with no indeterminate or failed sign. These audits are redundant: the exact rational proof objects remain the proof data used above.

9 Literature boundary and next problems

The global degree-exclusion theorems have to be separated from several adjacent results. Ghidelli proves fairness for $1 + ax^2 + x^5$ with an arbitrary nonnegative cofactor and no product-degree cap [1]. Consequently the $v = 5$ parts of Theorems 7.14 and 8.2 are reverifications in algebraic degrees 3 and 4, not new mathematical cases. Ghidelli's proof uses linear recurrences, nonvanishing resultants, and root asymptotics for that fixed exponent. The present parity-block expansion uses the arithmetic discreteness of the cubic and quartic catalogs to cover every odd exponent uniformly. Hare's algorithm instead scans candidate-factor coefficient patterns of factor degree at most 15 and eliminates all but 975 among 7,141,686 patterns [2]. That factor-degree classification and the present unbounded-exponent but fixed algebraic-degree theorems constrain different parameters, so neither computation should be described as subsuming the other.

There is also a direct integer-divisor literature. Drungilas, Jankauskas, and Šiurys give a finite-state algorithm deciding restricted-coefficient multiples for monic integer polynomials without unit-circle zeros and apply it to low-degree Borwein polynomials [5]. Idris and Sac-Épée study which integer polynomials divide Newman polynomials, including explicit nondivisibility and bounded-multiple computations [3]. Their subsequent work gives additional existence and nonexistence results for Newman and Littlewood multiples of certain integer polynomials [4]. These methods become relevant here after the conjugate product H_a converts a nonnegative real-factor question into an integer Newman-divisor question. The global cubic and quartic proofs instead return to the distinguished real embedding and certify the forced cofactor recurrence directly. The literature sources checked did not exhibit the parity-block certificate or an all-odd-v quartic exclusion, but that absence is a literature-search inference rather than a priority proof.

Up to reversal take $1 \le u \le v/2$. Theorem 3.2 resolves precisely the classes with $u \mid v$. For every remaining class, Theorem 3.3 removes $[1/2, 1)$ and Theorem 4.1 removes rational points. The norm sieve converts every fixed algebraic degree into a finite search.

Three directions remain concrete. First, Lemma 7.3 already gives an eventual negative coefficient for every real $a > 0$; to turn that into full fairness for the smaller gap two, push the six-block safe lattice beyond $j = 6v$ or rule out every nonstructural zero pair that could let a finite forced cofactor terminate with leading coefficient 1 before the first negative term. Second, determine whether the norm sieve and parity-block method remain computationally and structurally effective in algebraic degree 5. Third, organize propagation for smaller gaps $u > 2$ by residue classes or the continued fraction of v/u, and compare Theorem 4.2 systematically with general results on prescribed Newman divisors, especially the algorithms of Idris and Sac-Épée [3]. The current paper proves neither the full conjecture nor a counterexample.

AI research provenance

This manuscript results from a state-aware autonomous AI Researcher–Critic workflow. During each research iteration, the AI Researcher explored proofs, derived symbolic identities, searched for counterexamples, ran exact computations, implemented verification programs, surveyed relevant literature, and drafted the manuscript. In separate review iterations, the AI Critic checked claims, replayed computations, searched for gaps and edge cases, and returned structured feedback for the next Researcher iteration.

The human role was limited to initiating the project, defining the overall objective, configuring and supervising the workflow, selecting retained outputs, and authorizing release. The proof arguments, computational searches, verification code, critical assessments, and prose were generated primarily by the AI workflow and were not independently derived by the human orchestrator. The labels "Researcher" and "Critic" describe workflow roles rather than additional authors.

The accompanying repository provides the manuscript source, exact computational certificates, verification programs, and integrity manifests so that the workflow's mathematical outputs can be inspected and reproduced. The complete public workflow record, including the stable automation prompt, schedule, version chronology, and retained timing ledger, is provided in `docs/autonomous-research-workflow.md`.

Prompted workflow

The automation used a two-role, state-aware cycle. The Researcher prompt required at least 300 active minutes of proof work, exact computation, counterexample search, and literature checking, with exactly one incremented manuscript version created only after material progress. The Critic prompt required at least 60 active minutes of separate review, including computation replay, gap and edge-case searches, and severity-ranked feedback; the Critic did not edit the manuscript. Shared rules required hash validation before every role claim, prohibited both roles from running in one invocation, preserved earlier artifacts, and required selectable-text vector PDFs with visual inspection. The complete operational prompt appears in the workflow record cited above.

Schedule, iterations, and time accounting

Research artifacts span 10–28 July 2026. The workflow produced 18 numbered internal manuscript iterations; the released manuscript is intentionally unnumbered. From 17 July onward, the scheduler started one state-aware worker every six hours. Each invocation selected at most one role from the persisted phase. Researcher iterations required at least five active hours, Critic iterations at least one active hour, and every successful handoff ended the invocation.

The retained late-stage log records 40.68 hours of Researcher and Critic workflow time, including unsuccessful and repeated handoff attempts. The final production-and-review cycle comprised 358.38 Researcher minutes and 65.31 Critic minutes, or 7.06 hours in total.

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